

Researcher/Biostatistician applicant assignment

General instructions:

1. You may use any programming language (e.g., R, Python) to complete the task.
2. You may use the internet for research, but do not use ChatGPT or similar AI tools.
3. Please document your work process thoroughly and write clear, readable code. The result should allow a team member to understand your work and continue from where you left off if needed. In addition to performing the required analyses, your submission should include a clear interpretation of the results, a concise summary of findings, and a discussion of the rationale behind your choice of models and methods.
4. The time allotted for this task is up to **two days**. Please send back both the code and your answers by the deadline.

Question 1

Our researchers want to use medical records to investigate the risk of becoming diabetic among pre-diabetes patients.

Data were already extracted from the database and had preliminary pre-processing. You can find them in the file diabetes.csv.

Variables:

- *id* - subject identifier
- *age* - age at first drug use
- *male* - 0=female, 1=male
- *weight* - weight (kg)
- *height* - height (cm)
- *bmi* - body mass index= $\text{weight}/(\text{height}/100)^2$ (kg/m²)
- *futime* - days from diagnosis of pre-diabetes to diagnosis of diabetes or end of follow-up
- *status* - 0=not diabetic at last follow-up, 1=diabetic
- *type* - 0=treatment A, 1=treatment B

In the following questions, please use the provided dataset and perform the analysis whenever it is possible. If it is not possible, please describe the data, design, analysis, etc. needed to answer the question.

- a. In the population from which this dataset was derived, what is the overall risk of becoming diabetic within 5 years from pre-diabetes diagnosis? If possible, give both an estimate and a confidence interval.
- b. Which factors are associated with elevated risk?
- c. What would be the predicted 5-year risk of a person with the same characteristics as subject number 1?
- d. How good is this model? Is it good enough? How can you tell?

- e. Could you think of an alternative method to model this risk? If so, repeat c and d accordingly.
- f. Create visualizations to represent the data and results from the analyses above. Use plots such as histograms, scatter plots, survival curves, or any other relevant visualizations. Provide a brief interpretation of each visualization, explaining what it shows and how it supports your findings.
- g. Treatment B is a new type of medication that has entered the market a few years ago. The company claims it is better than the standard care (treatment A) in preventing diabetes. It is important for policy makers to know whether this claim is correct. Could you help investigate this claim?

Question 2

The American CDC wants to check the safety of the Bivalent booster vaccination for COVID-19. They reported the following method and design:

Analytical Strategy

We will tabulate the cumulative incidence of the targeted outcomes during pre-specified post-vaccination risk intervals. For each outcome, the cumulative number of events observed in the risk interval (1-21 days post vaccination) will be compared to the number expected as follows:

- *On each day, outcome incidence in the vaccinees who are in their risk interval (1-21 days post vaccination) will be compared to outcome incidence in vaccinees who are concurrently—on the same calendar date—in their comparison interval (22-42 days post vaccination).*
- *Analyses will be stratified in risk sets anchored to the calendar dates of outcome events.*
- *Analyses will be stratified by age group, sex, race, and clinic, as well as by calendar date.*

Statistical Analysis

Poisson regression will be used to compare outcome incidence during risk intervals with incidence during comparison intervals. The log of the expected count of events in a risk or comparison interval in a stratum on a calendar day will be modelled as a function of whether the stratum's vaccinees are in a risk versus a comparison interval on that day. The analysis will be conditioned on age-group, sex, race, and clinic, (which define the strata), as well as calendar day. The log person-days in each risk or comparison interval will be included in the regression model as the offset. This "offset" sets the expectation that if, for example, on Jan 20, 10% more vaccinees (in an age-sex-race-clinic stratum) are in their risk interval than in their comparison interval, then we expect—under the null—10% more outcome events in risk interval than in the comparison interval.

Rate ratios – estimating the ratio of outcome incidence in the risk interval divided by outcome incidence in the comparison interval – will be reported with 95% confidence intervals.

- a. Create an example of how the data table should look like in order to fit the CDC's proposed analysis. Then, create a plot to visualize the cumulative incidence of the targeted outcomes.

- b. What function and arguments would you use for modelling? Write the line of code that would use the data table from (a) to run the model.
- c. Could you suggest an alternative method to the one the CDC proposed? Repeat (a) and (b) accordingly.
- d. What are the pros and cons of using your suggestion, compared to the CDC's suggestion?